

Fazle Rabbi Rakib

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EDUCATION

Shahjalal University of Science and Technology

Bachelor of Science in Software Engineering

Jan 2018 – Dec 2022

Sylhet, Bangladesh

CGPA: 3.93/4.00

- Graduated with distinction, ranking 1st out of 50 students.
- **Relevant Coursework:** Data Structures and Algorithms, Software Requirement Engineering, Object Oriented Programming, Software Architecture and Design Patterns, Software Verification and Validation, Data Science, Artificial Intelligence, Machine Learning, Computer Data and Network Security.

RESEARCH EXPERIENCE

Undergraduate Research Supervisor

Software Engineering, IICT, Shahjalal University of Science and Technology

Mar 2024 – Present

- Mentored seven undergraduate research groups (four ongoing, three completed) in areas spanning AI, NLP, software engineering and LLMs.
- Supervise students through problem formulation, literature review, methodological design, modeling, evaluation, identifying and addressing limitations, and academic writing.

Research Assistant

Bengali.AI 🔗

Oct 2021 – Jan 2023

- Collaborated closely with a passionate team of Bengali NLP researchers worldwide who are enthusiasts in the open-source community.
- Led and supervised a team of 20+ annotators in developing "OOD-Speech: A Large Bengali Speech Recognition Dataset for Out-of-Distribution Benchmarking". Oversaw data collection and quality assurance to ensure dialectal and contextual diversity across Bengali-speaking regions. Contributed to designing annotation protocols and validation workflows, helping establish OOD-Speech as a benchmark dataset for robust and inclusive Bengali ASR research.
- Assisted in developing "BaDLAD: A Large Multi-Domain Bengali Document Layout Analysis Dataset" by collecting and curating document images from diverse real-world sources. Collaborated with the research team to ensure diversity and quality, contributing to the largest open-source Bengali dataset for document layout analysis and OCR research.

PUBLICATIONS

1. **FR Rakib**, SS Dip, S Alam, +11 authors. OOD-Speech: A Large Bengali Speech Recognition Dataset for Out-of-Distribution Benchmarking. Accepted in Interspeech 2023 (later withdrawn from proceedings), pages(879-883), August 2023. Preprint: arxiv.org/abs/2305.09688
2. MIH Shihab, MR Hasan, M Rahman, SM Hossen, MN Ansary, I Ahmed, **FR Rakib**, +10 authors. BaDLAD: A Large Multi-Domain Bengali Document Layout Analysis Dataset. In International Conference on Document Analysis and Recognition (ICDAR), pages 326-341, August 2023. DOI: https://doi.org/10.1007/978-3-031-41676-7_19.

RESEARCH IN PROGRESS

Integrating LLMs for Effective C/C++ Learning: Opportunities and Impacts

Ongoing | Code Generation, Prompt Engineering, LLM, Personalized Learning

Apr 2025 – Present

- Developed a classroom-deployed LLM-powered learning assistant supporting five instructional modes – General Question, Write Code, Fix Code, Explain Code, and Ask from Code – to provide structured context-aware guidance for students learning C/C++ programming. The system also offers three response formats: code, pseudo-code, and no-code, allowing learners to select the level of abstraction they need.
- Implemented dynamic prompt templates, in-app feedback (ratings & checklists), and structured issue reporting; currently deployed to a cohort of 60 students and used to iteratively refine prompts.

Bangla News Bias Detection Dataset with Hierarchical Annotation and Benchmarking *Sept 2025 – Present*
Ongoing | *Media Bias Detection, News Credibility, Bangla NLP, LLMs, AI*

- Research on formulating a structured news bias corpora, including dataset construction, annotation for stance and sentiment, and to guide the development of a Bangla news bias detection dataset and model pipeline for evaluating media framing and polarization patterns.
- A transformer-based model has been fine-tuned with the dataset and benchmarked with six LLMs, demonstrating that LLMs showed limited capability than the fine-tuned model in detecting news bias.

Evaluation of SCoT with different prompting strategies for Text-to-SQL generation *Oct 2025 – Present*
Ongoing | *NLP, Text-to-SQL, Databases, Prompting Strategies, LLMs*

- A comparative study of Structured Chain-of-Thought (SCoT), CoT, QDecomp, QDecomp_InnerCol and Standard prompting strategies for the Text-to-SQL generation task.
- Preliminary evaluation indicates that SCoT performed best due to its structured pass, candidate generation, and ranking pipeline, reducing the propagation of error compared to other methods for the specified task.

PROFESSIONAL EXPERIENCE

Lecturer, Software Engineering *Nov 2023 – Present*
Institute of Information and Communication Technology, Shahjalal University of Science and Technology Sylhet, Bangladesh

- Teaching a wide range of undergrad courses, including *Software Requirement Engineering, Object-Oriented Programming, Operating System and System Programming, Numerical Analysis, Web Technologies, Software Usability and Metrics, Software Verification and Validation, Information and Network Security, and Computer Graphics and Image Processing*.
- Integrating academic knowledge with industry experience to provide students with practical perspectives on software design, version control, clean coding, software architecture, and project presentation.
- Mentoring students through multiple project-based learning experiences — including structured programming, data structures, OOP with Java, and web application development — familiarizing them with industry-standard practices and techniques.
- Collaborating with students and colleagues on research initiatives.

Instructor, EDGE Project *Jul 2024 – Mar 2025*
Bangladesh Computer Council, ICT Division *Sylhet, Bangladesh*

- Conducted training on Python programming, Basic Web Development, and Foundational Data Science for 100+ undergraduate students from non-technical backgrounds under the EDGE (Enhancing Digital Government and Economy) project, funded by the World Bank and the Government of Bangladesh.

Guest Lecturer *Jan 2024 – Jun 2024*
Computer Science and Engineering, Sylhet Engineering College *Sylhet, Bangladesh*

- Instructed final-year engineering students in the *Digital Image Processing* course for one academic semester.

Software Engineer I *Mar 2023 – Dec 2023*
Pathao Limited *Dhaka, Bangladesh*

- Developed and maintained web portals for Pathao Pay, supporting secure digital payment workflows and customer-facing services.
- Collaborated in building and maintaining multiple internal dashboards to manage in-app notifications and ride operations.
- Collaborated in addressing vulnerabilities identified through third-party security audits by updating dependencies, reducing coupling, and improving software robustness.
- Designed and implemented reusable frontend components while collaborating with backend engineers, designers, QA engineers, and product managers.
- Contributed to maintainable software through modular architecture, version control, and clean coding practices.

Associate Engineer *Jun 2022 – Feb 2023*
Pathao Limited *Dhaka, Bangladesh*

- Developed reusable internal UI components that improved consistency and reduced duplicated implementation across multiple applications.
- Collaborated with the backend team to build responsive web interfaces.
- Participated in sprint planning, implementation, testing, and feature delivery in a collaborative Agile environment.

Intern

Pathao Limited

Sep 2021 – May 2022

Dhaka, Bangladesh

- Completed structured training in modern frontend development, using HTML, CSS, JavaScript, and Vue.js, while developing web-based applications including a To-Do List and a Base64 Encoder & Decoder tool.
- Contributed by implementing basic features and validation for an Access Control Dashboard.
- Gained practical experience working with large-scale production codebases, Git workflows, code reviews, and collaborative software development practices.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, C/C++, Bash, Java, SQL, HTML/CSS
- **Technologies/Frameworks:** VueJS, Node.js, Express, React, Angular, Tailwind CSS
- **Developer Tools:** Git, GitHub, GitLab, Docker, VS Code, IntelliJ, Jupyter Notebook, Postman
- **Machine Learning & AI:** LLMs, Pandas, NumPy, Matplotlib, Scikit-Learn
- **Other tools & Technologies:** Chrome Developer Tools, Figma, Jira, Trello

PROJECTS

Base64 Converter | *VueJS, Vue-Router, Crypto Tools, Base64, RSA*

Nov 2022 – Dec 2022

- Created a web-based utility enabling users to encode or decode Base64 data. I implemented advanced features, including RSA key-pair generation (optional name/email/passphrase, 2048-/4096-bit), message-digest calculations across multiple hashing algorithms (MD4, MD5, SHA-256, SHA-512), and integrated built-in packages (bcrypt, cryptico, hash-wasm).
- Built a clean, tabular UI for results and deployed the static site on Netlify .

Department Collaboration System | *NodeJS, Express, Angular, Sequelize, Angular Material*

Jul 2021 – Dec 2021

- Designed and developed an online collaboration system to assist with departmental activities (both students & teachers) in the COVID-19 pandemic.
- Contributed to developing a modular backend architecture using well-structured RESTful APIs that ensured access control and data management. Implemented key modules for user authentication, course and classroom management, content interaction, and class routine generation.

FTP-Server | *Java, Java FX, Socket programming*

Jan 2021 – Jun 2021

- Developed a Java-based client-server file transfer system, part of a lab task on the Computer Networking course lab, using Java FX for the GUI and Java sockets for network communication.
- Implemented core logic for both client and server connections, enabled file upload/download operations, stored file metadata (name, size, upload time), and supported file deletion and renaming. Conducted multi-scenario testing and fixed bugs to ensure reliable performance.

E-Provost | *Laravel, MySQL, Bootstrap, Docker*

Jul 2020 – Dec 2020

- Developed a role-based academic management platform using Laravel to assist and automate the daily operations of university hall management, including student concerns, hall announcements, and administrative workflows. Implemented user authentication, dynamic routing, and database-driven dashboards to manage operations efficiently.

OPEN SOURCE CONTRIBUTION

VS Code Configuration for Competitive Programming | *Repository: jspw/VS-Code-Config*

Oct 2024

- Contributed to an open-source project that provides Visual Studio Code workspace and task configurations to streamline competitive programming workflows, enabling easy input/output and faster test cycles.
- Updated configuration scripts, refining documentation, and cleaning up redundant settings to enhance default PowerShell compatibility for Windows users.

HONORS AND AWARDS

Judge of Hackathon | *SUST CSE Carnival 2024*

Feb 2024

Runners Up in Faculty Cricket Tournament | *SUST*

Jan 2024

University Merit Stipend(Twice) | *SUST*

Jan 2022, July 2022

Ranked among the Top 5 teams in the Hackathon | *Leading University, Sylhet*

2022

Runners Up in Inter-department Table Tennis Tournament | *SUST*

2022

Codeforces Specialist | *Max Rating: 1424*

2022

Competitive Programming Problem Solving | *Count: 700+*

2018 - 2022

Ranked among the Top 7 teams in the HULT prize | *SUST*

2019

VOLUNTEER EXPERIENCE

Event Coordinator

SUST SWE Technovent 2023

Oct 2022 – Feb 2023

Sylhet, Bangladesh

- Led one of the core teams in SWE Technovent, a national tech event with 400+ participants, supporting coordination, logistics, and event hosting.
- Organized and managed departmental Sports Week involving all student batches in various indoor and outdoor games.

Event Organizer

IICT Directors' Sports Week 2022

Aug 2022 – Sep 2022

Sylhet, Bangladesh

- Organized and coordinated event plans, established game and disciplinary rules, and led volunteers.

Executive Member

SWE Society, SUST

Jan 2020 – Dec 2020

Sylhet, Bangladesh

- Organized and managed various yearly co-curricular and extra-curricular events.

REFERENCES

Dr. Mohammad Shahidur Rahman

Professor

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Dr. Ahsan Habib

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